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ALARM CONTROL DEVICE

Abstract

The disclosure relates to an alarm control device. The alarm control device does not activate the alarm device when the door is unlocked without using the authenticated key from the interior of the vehicle while the alarm device is in the valid state and the authenticated key is in the interior of the vehicle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to Japanese Patent Application No. 2024-028600 filed on Feb. 28, 2024, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to an alarm control device.

2. Description of Related Art

[0003] Japanese Unexamined Patent Application Publication No. 61-237778 (JP 61-237778 A) describes an alarm control device that activates an alarm when a door of a vehicle is unlocked without using a remote key after the door is locked using the remote key.

SUMMARY

[0004] The alarm control device described in JP 61-237778 A activates an alarm device even when the door of the vehicle is unlocked using a door key in a state in which a user of the vehicle possesses the remote key after the door of the vehicle is locked using the remote key.

[0005] An aspect of the present disclosure provides an alarm control device that controls an alarm device that activates an alarm when a door of a vehicle is unlocked, the alarm control device controlling the alarm device so as to shift to an activated state when the door is locked using an authenticated key, and so as to shift to an inactivated state when the door is unlocked using the authenticated key, in which [0006] the alarm device is not caused to issue an alarm when the door is unlocked from an inside of a cabin of the vehicle without using the authenticated key in a state in which the alarm device is in the activated state and the authenticated key is in the inside of the cabin of the vehicle.

[0007] It is possible to issue an alarm in a preferable manner.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

[0009] FIG. 1 is a block diagram illustrating an electrical configuration of an alarm control device;

[0010] FIG. 2 is a flow chart showing a flow of an alarm control process at the time of unlocking; and

[0011] FIG. 3 is a flowchart illustrating a flow of unlocking method determination processing.

DETAILED DESCRIPTION OF EMBODIMENTS

First Embodiment

[0012] Hereinafter, a first embodiment of the alarm control device will be described with reference to FIGS. 1 to 3.

[0013] The alarm control device for controlling the alarm device includes a processor, a storage, and a memory. A program for controlling the alarm device is stored in the storage. The processor controls the alarm device by executing a program based on information stored in the memory.

Electrical Configuration of Alarm Control Device

[0014] As shown in FIG. 1, the alarm control device **10** is electrically connected to the alarm device **11**. The alarm device **11** is a device that activates an alarm. For example, the alarm is a buzzer ringing. The alarm device **11** may transition between a valid state in which an alarm can be activated and a disabled state in which an alarm cannot be activated. The alarm control device **10** can shift the alarm device **11** from the disabled state to the enabled state. The alarm control device **10** can shift the alarm device **11** from the enabled state to the disabled state. The alarm control device **10** can activate an alarm to the alarm device **11**.

[0015] The alarm control device **10** is electrically connected to the door **12** of the vehicle. The alarm control device **10** can acquire, from the door **12**, an opening amount signal capable of recognizing an opening amount of a window provided in the door **12**. For example, the opening amount signal is output from the door **12** when the opening amount of the window of the door **12** changes. The window of the door **12** is opened and closed by a motor provided in the door **12**. The motor of the door **12** is operated by operating an opening/closing switch provided in the door **12**. The alarm control device **10** can acquire, from the door **12**, a locking signal capable of recognizing whether or not the door **12** is locked. Locking and unlocking the door **12** is to lock and unlock the door lock provided on the door **12**. In the present specification, when “unlock” is described, the door **12** is unlocked by a normal method. On the other hand, the phrase “unlocked” includes a state in which the door **12** is unlocked regardless of the proper method. The door **12** is locked and unlocked by operating a locking and unlocking switch provided in the door **12**. For example, the locking and unlocking switch is provided in the door **12** so as to be operable from the interior of the vehicle. Locking and unlocking the door **12** by operating a locking and unlocking switch from the interior of the vehicle is to lock and unlock the door **12** by a regular method.

[0016] The alarm control device **10** is electrically connected to an authenticated key **13** for locking and unlocking the door **12**. The authenticated key **13** is a key that has been authenticated by the verification control device provided in the vehicle. For example, the verification control device executes ID verification of the authenticated key **13** by a narrow-area radio (communication range: several meters). The authenticated key **13** can lock and unlock the door **12** by narrow-area radio. Locking and unlocking the door **12** using the authenticated key **13** is locking and unlocking the door **12** in a regular manner.

[0017] For example, the alarm control device **10** can acquire ID verification from the verification control device. The alarm control device **10** can identify the authenticated key **13** based on ID verification obtained from the verification control device. The alarm control device **10** can acquire the periodic signal by the narrow range radio from the authenticated key **13**. For example, the authenticated key **13** outputs a periodic signal at every specified period. For example, the alarm control device **10** is configured to include an antenna for acquiring a signal by narrow-area radio from the authenticated key **13**. For example, the alarm control device **10** can acquire a signal by narrow-area radio from the authenticated key **13** when the authenticated key **13** is present in the interior of the vehicle.

[0018] The alarm control device **10** is electrically connected to the camera **14**. The camera **14** can image the surroundings of the vehicle. The alarm control device **10** can acquire an image of the surroundings of the vehicle captured by using the camera **14** by acquiring an image signal from the camera **14**. For example, the camera **14** is provided in the interior of the vehicle. That is, the peripheral image is an image through the window of the door **12**.

[0019] Processing executed by the alarm control device when the door is locked or unlocked.

[0020] Processing executed by the alarm control device **10** when the door **12** is locked or unlocked will be described.

[0021] First, processing when the door **12** is locked and unlocked using the authenticated key **13** will be described.

[0022] The alarm control device **10** shifts the alarm device **11** to the valid state when the door **12** is locked using the authenticated key **13**. For example, the alarm control device **10** determines whether or not the door **12** is locked using the authenticated key **13** based on the lock signal. The alarm control device **10** shifts the alarm device **11** to the invalid state when the door **12** is unlocked using the authenticated key **13**. For example, the alarm control device **10** determines whether or not the door **12** is unlocked using the authenticated key **13** based on the lock signal.

[0023] Processing when the door **12** is locked and unlocked without using the authenticated key **13** will be described.

[0024] The alarm control device **10** does not shift the alarm device **11** to the valid state when the

door **12** is locked without using the authenticated key **13**. The door **12** can be locked without using the authenticated key **13** by operating the locking and unlocking switch of the door **12**. When the door **12** is unlocked without using the authenticated key **13**, the alarm control device **10** does not activate the alarm when the alarm device **11** is in an invalid state.

[0025] An unlocking alarm control process executed when the alarm device **11** is in the valid state when the door **12** is unlocked without using the authenticated key **13** will be described.

[0026] As illustrated in FIG. 2, in the unlocking time alarm control process, the alarm control device **10** determines whether or not there is an authenticated key **13** in the interior of the vehicle (**S10**). For example, the alarm control device **10** determines that the authenticated key **13** is present in the interior of the vehicle when the periodic signal from the authenticated key **13** is acquired.

[0027] When the authenticated key **13** is present in the interior of the vehicle (**S10**: YES), the alarm control device **10** executes the unlocking-method determination process. The unlocking method determination process is a process for determining whether the door **12** has been unlocked from the interior of the vehicle or from the exterior of the vehicle.

[0028] As illustrated in FIG. 3, in the unlocking-method determination process, the alarm control device **10** determines whether or not the opening amount of the windows of the doors **12** is less than the predetermined amount based on the opening amount signal (**S20**). The alarm control device **10** determines that the opening amount of the window of the door **12** is less than the predetermined amount when the window of the door **12** is not opened. On the other hand, when the window of the door **12** is opened, the alarm control device **10** determines that the opening amount of the window of the door **12** is not less than the predetermined amount. For example, the predetermined amount may be an opening amount into which an instrument for operating the locking and unlocking switch of the door **12** can be inserted from the outside of the vehicle. Locking and unlocking the door **12** by inserting the tool from the outside of the vehicle and operating the locking and unlocking switch from the outside of the vehicle is not the locking and unlocking of the door **12** by a regular method.

[0029] When the opening amount of the windows of the door **12** is less than the predetermined amount (**S20**: YES), the alarm control device **10** acquires peripheral images of the vehicles captured by using the camera **14** (**S21**). The alarm control device **10** determines whether or not the surrounding images are normal (**S22**). For example, the alarm control device **10** determines whether or not the surrounding image is normal using a learned image recognition model that has been previously machine-learned. The alarm control device **10** determines, based on the image recognition model, whether or not the door **12** has been unlocked regardless of the normal method from the surrounding image. Then, based on the determination result, it is determined whether or not the surrounding image is normal. For example, the alarm control device **10** determines that the surrounding image is not normal when it is determined that the door **12** is unlocked from the outside of the vehicle based on the surrounding image.

[0030] When the surrounding images are normal (**S22**: YES), the alarm control device **10** determines that the door **12** is unlocked from the interior of the vehicle, and ends the unlocking-method determination process. On the other hand, when the opening amount of the window of the door **12** is equal to or greater than the predetermined amount (**S20**: NO) or when the surrounding images are not normal (**S22**: NO), the alarm control device **10** determines that the door **12** is unlocked from the outside of the vehicle, and ends the unlocking-method determination process.

[0031] Returning to the explanation of the unlocking-time alarm control process, as shown in FIG. 2, the alarm control device **10** determines whether or not the door **12** is unlocked from the interior of the vehicle as a result of the unlocking-method determination process (**S12**). When it is determined that the door **12** is unlocked from the interior of the vehicle (**S12**: YES), the alarm control device **10** shifts the alarm device **11** to the disabled state. On the other hand, when the authenticated key **13** is not present in the interior of the vehicle (**S10**: NO), and when it is determined that the door **12** is unlocked from the exterior of the vehicle (**S12**: NO), the alarm

control device **10** causes the alarm device **11** to activate an alarm.

Operations of Present Embodiment

[0032] The operation of the present embodiment will be described.

[0033] The alarm control device **10** shifts the alarm device **11** to the valid state when the door **12** is locked by using the authenticated key **13**. Further, the alarm control device **10** shifts the alarm device **11** to the invalid state when the door **12** is unlocked by using the authenticated key **13**. The alarm control device **10** activates the alarm device **11** when the alarm device **11** is in a valid state and the door **12** is unlocked without using the authenticated key **13** in a state (S10: NO) in which the authenticated key **13** is not in the interior of the vehicle (S14). In addition, in the alarm control device **10**, there is a case in which the alarm device **11** is in the valid state and the authenticated key **13** is in the interior of the vehicle (S10: YES), and the door **12** is unlocked from the outside of the vehicle without using the authenticated key **13** (S12: NO). In this instance, the alarm device **11** is activated (S14).

[0034] On the other hand, in the alarm control device **10**, when the alarm device **11** is in the valid state and the authenticated key **13** is in the interior of the vehicle (S10: YES), the door **12** is unlocked from the interior of the vehicle (S12: YES). In this case, the alarm device **11** is switched to the disabled state (S13). That is, the alarm control device **10** does not activate the alarm device **11** when the door **12** is unlocked from the inside of the vehicle in a state (S10: YES) in which the alarm device **11** is in the valid state and the authenticated key **13** is in the inside of the vehicle (S12: YES).

[0035] When the door **12** is unlocked while the window of the vehicle is opened by a predetermined amount or more (S20: NO), the alarm control device **10** determines that the door **12** is unlocked from the outside of the vehicle without using the authenticated key **13** (S24). When the door **12** is unlocked, the alarm control device **10** acquires images of the surroundings of the vehicle (S21). Based on the images, the alarm control device **10** determines whether or not the door **12** is unlocked without using the authenticated key **13** from the outside of the vehicle (S22).

Effects of Present Embodiment

[0036] The effects of the present embodiment will be described.

[0037] (1) By locking the door **12** using the authenticated key **13**, after the alarm device **11** is activated, if the door **12** is unlocked from the interior of the vehicle without the authenticated key **13** in the presence of the authenticated key **13** in the interior of the vehicle, the alarm is not activated. For example, when the user takes a nap in the interior of the vehicle, it is conceivable to lock the door **12** for security. Further, for security, it is conceivable to enable the alarm device **11** by locking the door **12** using the authenticated key **13**. In such a situation, when the user unlocks the door **12** by operating the locking/unlocking switch of the door **12**, the user does not feel troublesomeness by not operating the alarm to the alarm device **11**. That is, the alarm can be activated suitably.

[0038] (2) On the other hand, when the door **12** is unlocked from the outside of the vehicle even in a state where the authenticated key **13** is present in the interior of the vehicle, an alarm is activated to the alarm device **11**. For example, in the above-described situation, when the door **12** is unlocked from the outside of the vehicle, the security can be improved by activating an alarm to the alarm device **11**. That is, the alarm can be activated suitably.

[0039] (3) For example, when the window of the door **12** is opened by a predetermined amount or more, the locking/unlocking switch of the door **12** can be operated by inserting an instrument or a hand through the opening. Therefore, in such a situation, it is determined that the door **12** is unlocked without using the authenticated key **13** from the outside of the vehicle, and the alarm is activated to the alarm device **11**, whereby the security can be improved. That is, the alarm can be activated suitably.

[0040] (4) For example, when an attempt is made to unlock the door **12** from the outside of the vehicle without using the authenticated key **13**, the situation is included in an image obtained by

capturing the surroundings of the vehicle. Therefore, in a case where the door **12** is unlocked, it is determined whether or not the door **12** is unlocked from the outside of the vehicle without using the authenticated key **13** based on the image, whereby the security can be improved while suppressing the user from feeling troublesomeness. That is, the alarm can be activated suitably.

Modifications

[0041] The above embodiment may be modified as follows. Note that the above-described embodiments and the following examples may be combined with each other within a range not technically inconsistent with each other. [0042] The alarm control device **10** may determine that the door **12** is unlocked from the interior of the vehicle when the door **12** is unlocked without using the authenticated key **13** while the authenticated key **13** is in the interior of the vehicle. That is, the alarm control device **10** may not activate the alarm device **11** when the door **12** is unlocked without using the authenticated key **13** while the authenticated key **13** is in the interior of the vehicle. According to this, it is possible to suppress the control of the alarm control device **10** becoming complicated. [0043] The alarm control device **10** may execute **S21** process without executing **S20** process in the unlocking process determination process. In other words, the alarm control device **10** does not need to determine whether or not the door **12** is unlocked from the interior of the vehicle based on the opening amount of the window of the door **12**. According to this, it is possible to suppress the control of the alarm control device **10** becoming complicated. [0044] The alarm control device **10** may execute **S23** process without executing **S22** process in the unlocking process determination process. In other words, the alarm control device **10** does not need to determine whether or not the door **12** is unlocked from the interior of the vehicle based on the peripheral image of the vehicle. According to this, it is possible to suppress the control of the alarm control device **10** becoming complicated. [0045] The alarm control device **10** may be capable of determining that the door **12** is unlocked from the outside of the vehicle by a method in addition to or instead of the method described in the above embodiment. [0046] The alarm control device **10** may terminate the unlocking alarm control process without executing **S13** process in the unlocking alarm control process. In other words, even when the door **12** is unlocked by a regular method, the alarm control device **10** may not activate the alarm while maintaining the alarm device **11** in the valid state when it is performed without using the authenticated key **13**. For example, in a situation where the user takes a nap in the interior of the vehicle, the alarm is not activated for a certain period of time after the user temporarily unlocks the door **12**. In this case, the alarm device **11** is activated after a predetermined period of time has elapsed, so that the security can be improved. That is, the alarm can be activated suitably. [0047] The door **12** may be locked and unlocked using a different authenticated key than the authenticated key **13**. For example, the authenticated key different from the authenticated key **13** is a spare key different in ID from the authenticated key **13**. In this case, the alarm control device **10** shifts the alarm device **11** to the valid state when the door **12** is locked by using the spare key. In addition, the alarm control device **10** shifts the alarm device **11** to the invalid state when the door **12** is unlocked by using the spare key. For example, the alarm control device **10** may make an affirmative determination when at least one of the authenticated key **13** and the spare key is in the interior of the vehicle in **S10** of the unlocking alarm control process. For example, the alarm control device **10** may make an affirmative determination when, in **S10** of the unlocking alarm control process, the key used to shift the alarm device **11** to the valid status among the authenticated key **13** and the spare key is in the interior of the vehicle. In this case, the alarm control device **10** may make an affirmative determination even when the door **12** is unlocked by a key that differs from the key used when the alarm device **11** is shifted to the valid state **S12** the unlocking alarm control process.

Claims

1. An alarm control device that controls an alarm device that activates an alarm when a door of a vehicle is unlocked, the alarm control device controlling the alarm device so as to shift to an activated state when the door is locked using an authenticated key, and so as to shift to an inactivated state when the door is unlocked using the authenticated key, wherein the alarm device is not caused to issue an alarm when the door is unlocked from an inside of a cabin of the vehicle without using the authenticated key in a state in which the alarm device is in the activated state and the authenticated key is in the inside of the cabin of the vehicle.
 2. The alarm control device according to claim 1, wherein the alarm device is caused to issue an alarm when the door is unlocked from an outside of the cabin of the vehicle without using the authenticated key when the alarm device is in the activated state.
 3. The alarm control device according to claim 2, wherein it is determined that the door is unlocked from the outside of the cabin of the vehicle without using the authenticated key when the door is unlocked in a state in which a window of the vehicle is opened by a predetermined amount or more.
 4. The alarm control device according to claim 2, wherein: an image that captures surroundings of the vehicle is acquired when the door is unlocked; and it is determined based on the image whether the door is unlocked from the outside of the cabin of the vehicle without using the authenticated key.
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